

Cesarean delivery in public and private facilities

Research update | 3 October 2026 | NFHS-4 and NFHS-5

Following your recommendations, we narrowed the facility comparison and completed cross-wave, subgroup, policy-period and historical state-context analyses. Below we summarise the revisions, results and decisions on which we would value your guidance.

Revised scope and interpretation

Your recommendation	What we did
Use narrower facility categories.	Reconstructed the primary comparison from raw m15=21 (government/municipal hospital) versus m15=31 (private hospital/maternity home/clinic). Eligible births have valid outcome/weight and recall under 60 months.
Keep the aim associational; do not highlight unsupported causal estimates.	Observed survey-weighted proportions lead; model-adjusted associations are additional. Old AIPW/TMLE headlines and E-values are omitted. Estimator agreement does not establish causal identification.
Clarify comparisons and label first delivery appropriately.	Checked raw variable definitions and harmonised geography/reference populations; examined residence/wealth groups. First delivery is a subgroup, retaining first-delivery twins. Planning and previous experience remain possible, untested explanations.

National results for the narrow cohorts

Wave / cohort	Births	Public %	Private %	Observed gap [95% CI]	Adjusted gap [95% CI]
NFHS-4 All births	114,263	18.67	41.08	22.42 [21.41, 23.42]	19.17 [18.20, 20.13]
NFHS-4 First delivery	52,851	21.88	45.50	23.62 [22.24, 24.99]	21.21 [19.83, 22.59]
NFHS-5 All births	115,197	22.38	47.54	25.16 [24.27, 26.04]	23.76 [22.85, 24.67]
NFHS-5 First delivery	51,854	26.15	51.77	25.62 [24.38, 26.85]	25.24 [23.95, 26.53]

Throughout: gaps are private minus public, in percentage points (pp). CIs are approximate marginal 95% Taylor t intervals, primary singleton rule adjust, unless stated otherwise. National models use each wave/cohort's own reference; cross-wave results below use a shared reference.

The fixed-ridge models (0.00001) adjust for state, wealth, residence, schooling, religion, social group and plurality, plus birth order for all births. Date/recall checks excluded 979 NFHS-5 narrow births and none in NFHS-4. Clinical indication, prior cesarean history, referral, emergency status and facility capability remain incompletely measured.

SHAP remains predictive attribution only. Earlier weak, uneven prediction-score groups retain their original separate scope; they are not rerun narrow-cohort or reliable individual-effect findings. Fold stability remains an implementation check, not a headline result. The required causal conditions are not justified; we do not claim every assumption was empirically proved to fail.

Cross-wave and subgroup comparisons

Both wave models use the **same equal-wave pooled reference** in 33 comparable states/UTs. Raw state labels were harmonised; boundary-changing units were excluded. Wealth quintiles remain relative ranks, and interview-time covariates are not certified birth-time values.

Shared-reference sector gaps

Population	Reference births	NFHS-4 gap [95% CI]	NFHS-5 gap [95% CI]	Wave change [95% CI]
All narrow births	218,632	19.05 [18.09, 20.00]	23.71 [22.78, 24.64]	+4.66 [3.33, 5.99]
First delivery	99,802	21.01 [19.62, 22.40]	25.01 [23.67, 26.34]	+4.00 [2.08, 5.92]

Training n: 109,006/110,595 for all births and 50,527/49,799 for first delivery. Geography, interactions and reference differ from the national analysis. Primary reference cells contain at least one birth per sector in each wave. With at least 30, wave changes are +4.63 [3.32, 5.94] and +4.19 [2.33, 6.04] pp; first-delivery exclusions rise to about 15-17% of retained-geography weight. Support stability does not certify full joint overlap.

Formal changes in residence and wealth heterogeneity

Private-sector interactions allow subgroup gaps to differ. Each contrast below tests whether that between-group gap difference changed from NFHS-4 to NFHS-5. References are shared across waves within a subgroup, but differ across subgroups.

Subgroup-gap contrast	All births: change [95% CI]	Holm p	First delivery: change [95% CI]	Holm p
Rural versus urban	+3.57 [0.82, 6.31]	0.076	+3.05 [-0.91, 7.00]	0.597
Wealth quintile 2 versus 1	-2.13 [-6.01, 1.75]	0.597	-4.44 [-10.24, 1.36]	0.597
Wealth quintile 3 versus 1	-3.88 [-7.85, 0.10]	0.336	-3.85 [-9.65, 1.94]	0.597
Wealth quintile 4 versus 1	-6.80 [-10.67, -2.92]	0.0059	-7.96 [-13.70, -2.22]	0.059
Wealth quintile 5 versus 1	-5.71 [-9.97, -1.44]	0.070	-4.96 [-11.21, 1.28]	0.597

Holm adjusts the ten tests together; only all-birth quintile 4 versus 1 meets 0.05. The same rejection pattern holds in the separate stricter-support family. This added correction is exploratory, not prospectively specified; CIs remain unadjusted marginal intervals. It does not adjust other analyses. NFHS-4 wealth gaps are not monotonic; no universal gradient or causal group effect is claimed.

Training-population sensitivity on one first-delivery reference

NFHS-4: all-birth-trained gap 19.61 versus first-delivery-trained 21.01; difference **+1.40 [0.47, 2.32]** pp.

NFHS-5: 24.02 versus 25.01; **+0.98 [0.11, 1.86]** pp.

Both fits omit birth order and use matching predictors/reference; joint covariance accounts for overlapping training samples. Stricter-support differences are +1.45 and +1.04 pp. This is a training sensitivity, not a birth-population or causal contrast. National support checks changed gaps by at most 0.30 pp.

Policy periods and historical state context

We added birth-period and historical state-context analyses as requested. PMSMA (2016), LaQshya (December 2017) and SUMAN (October 2019) describe the policy environment; maternal mortality goals are not cesarean-rate targets.

Five-period comparison on one common-state reference

Birth period	Adjusted gap, pp	Approximate 95% CI
Before January 2016	21.32	[17.79, 24.84]
January-November 2017	23.68	[21.36, 26.01]
January 2018-September 2019	23.71	[21.95, 25.48]
November 2019-February 2020	25.84	[22.05, 29.63]
March 2020 onward	24.36	[20.89, 27.84]

Same 11 states; common reference n=42,047; excluded target weight 0.671%. Transition year 2016 and launch months are omitted. All five period-change CIs include zero; latest minus earliest: +3.05 [-1.89, 7.99] pp. This does not establish equivalence or policy effects; the final period also coincides with pandemic disruption.

Paired historical-context sensitivity and coverage

Wave / context	Excluded weight: all / first	Gap change: all births	Gap change: first delivery
NFHS-4 / Actual spending	10.4 / 10.4%	+0.0006	+0.0013
NFHS-4 / Actual + planned budget	65.5 / 65.1%	-0.0010	+0.0220
NFHS-4 / State milestones	0.3 / 0.3%	+0.0047	-0.0046
NFHS-5 / Actual spending	3.4 / 3.5%	-0.0003	-0.0001
NFHS-5 / Actual + planned budget	36.1 / 36.6%	+0.0022	+0.0020
NFHS-5 / Actual + CHC, 12m	62.1 / 62.5%	-0.0053	+0.0049
NFHS-5 / Actual + CHC, 24m	48.6 / 48.8%	-0.0051	-0.0031

All 14 paired change CIs include zero; largest: +0.0220 [-0.0337, 0.0777] pp. Models use identical linked references and calendar adjustment. RBI shares use the preceding financial year; CHC snapshots precede births. Milestones cover West Bengal medicine shops/Maharashtra JSSK. NFHS-4 facility cases lack linked births; NFHS-5 policy terms are absorbed.

West Bengal exclusion: national narrow-cohort sensitivity

Wave	All births: gap / change [95% CI]	First delivery: gap / change [95% CI]
NFHS-4	Gap 17.56; change -1.61 [-1.96, -1.26]	Gap 18.78; change -2.43 [-2.95, -1.90]
NFHS-5	Gap 22.03; change -1.73 [-2.01, -1.45]	Gap 23.27; change -1.97 [-2.45, -1.49]

State exclusion lowers the gap but leaves it positive. State/calendar terms absorb 89-96% of unweighted finance/facility variance (partial sector $|r| < 0.016$). Prioritise better-covered spending results; restricted analyses are supplementary. Context enters as macro main effects, so tiny changes do not establish no context effect or test sector-gap modification.

Computational feasibility, limitations and guidance

Following your encouragement, we extended the federated pilot to state-adjusted fitting, repeated penalty validation and approximate survey uncertainty. It establishes local computational feasibility; remote deployment and privacy validation remain a possible separate methods project.

Component	Completed work / result
Local fitting and penalty	115,197 births; 36 state/UT partitions; four worker processes. State indicators are included. Ridge 0.00001 was selected across five inner PSU-validation splits in the original outer training set; nearby penalties performed similarly.
Full-refit agreement	Gap 23.76 pp [22.85, 24.67] ; independent pooled fitting agrees within 0.000001 pp. The outer test was previously used and is a reused diagnostic, not new independent validation.
Privacy and novelty	Trusted preparation/pooled validation read private rows locally; worker aggregates are unencrypted. No secure aggregation, differential privacy or remote deployment is implemented. No first-ever NFHS novelty claim is made without a literature review.

Qualifications retained in the revised interpretation

Issue	What remains unresolved
Survey design	Joint fitted-model/reference Taylor uncertainty uses state-stratum-PSU keys and observed zero-domain PSUs; primary singleton centering has sensitivity checks. It is a with-replacement approximation, without FPC/lower-stage variance. NFHS-4 household roster is unavailable.
NFHS-5 roster discrepancy	All nine tested keys yield 30,170 observed units, versus 30,198 published completed-fieldwork PSUs. The 636,699 interviewed households match the national total; this does not explain the 28-unit discrepancy. Roster completeness and its variance effect remain unresolved.
Model and context scope	Preprocessing, penalties, support and historical sources are fixed; period fits condition on profiling 23 separated rows absent from the reference. CHC staffing covers rural public facilities; state is interview residence. Milestones do not measure uptake or cover every policy. Birth-level variance is not state-policy inference.

Private records remain local; reviewed aggregates do not certify causal identification or full design variance. Sources: IIPS/ICF survey counts, NHM programmes, RBI finance, CBHI National Health Profiles 2018/2019, and official West Bengal/Maharashtra milestones. Extraction details remain in supporting records.

Questions on which we would value your guidance

Decision	Question
Publication uncertainty	Are the current conditional Taylor intervals and roster/singleton qualifications adequate, or would you recommend further survey-design review or PSU resampling with model refitting before publication?
State-context presentation	Would you favour actual spending as the main contextual sensitivity, with the more restricted budget/facility analyses supplementary?
Federated development	Should we retain the local prototype, or pursue remote deployment and validated privacy protection as a separate methods contribution? What would you prioritise?
Earlier deferred guidance	We have omitted E-values as advised. Following your earlier note that you would consider the sensitivity question, is any further associational sensitivity needed beyond the completed support, state-exclusion and subgroup checks?